

APEX: AI Platform of Platforms

Cloud-Native AI Agent Architecture | TOGAF 10 ADM + AI-DLC

The authoritative Enterprise Architecture blueprint for the APEX programme at GlobalCorp Enterprises. Applies TOGAF 10 ADM as governance skeleton and AI-DLC as velocity engine to create a cloud-native AI Platform of Platforms.

Covers all nine ADM phases, full team structure, RACI, responsibility mappings, cloud-native service design, and regulatory compliance across EU AI Act, DORA, GDPR, MiFID II, and internal model risk policy.

ORGANISATION

GlobalCorp Enterprises

FRAMEWORK

TOGAF 10 ADM + AI-DLC

DOCUMENT REFERENCE

EA-APEX-MASTER-001

CLASSIFICATION

CONFIDENTIAL

PROGRAMME

APEX — AI Platform of Platforms

ARCHITECTURE APPROACH

Cloud-Native / Vendor-Agnostic

ISSUE DATE

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STATUS

APPROVED — FINAL

OVERVIEW

Executive Summary

Programme context, goals, and key metrics

Programme Purpose

APEX is GlobalCorp's strategic programme to replace 23 siloed AI tools with a unified, governed **AI Platform of Platforms** built on cloud-native open standards. It applies the TOGAF 10 Architecture Development Method (ADM) as its governance framework and the AWS AI-Driven Development Lifecycle (AI-DLC) as its delivery methodology — positioning AI as a central collaborator across design, build, and operations, not merely a peripheral productivity tool.

The platform provides every GlobalCorp division with reusable AI agent capabilities, governed knowledge bases, shared observability, and a self-service marketplace — eliminating the time, cost, and compliance risk of building isolated AI systems.



What This Programme Delivers

APEX delivers three interlocking outcomes:

- Velocity** — time-to-market for AI use cases reduced from 9.2 months to a planning baseline of 5–6 weeks through AI-DLC Construction phase, reusable agent templates, and self-service onboarding
- Governance** — five-tier continuous governance model replacing quarterly gate reviews; automated conformance, real-time observability, and Decision Explanation Artefacts for every regulated agent decision
- Platform Economics** — \$14M annual duplicated spend eliminated through shared infrastructure, reusable tooling, and centralised compliance — generating positive NPV within 24 months of platform go-live

Programme Pain Chain — Current State vs. Target

Problem Area	Current State	APEX Target	Delivery Mechanism
Time-to-market for AI use cases	9.2 months	5–6 weeks (planning baseline)	AI-DLC Construction; reusable templates; automated compliance pipeline
Duplicated AI tool spend	\$14 M / year	\$3 M / year	Platform consolidation; shared observability; agent marketplace
Regulatory governance gaps	47 open gaps	0 gaps	Automated conformance; DEA pattern; DORA change framing
Developer satisfaction (NPS)	–12	Target: +55	Self-service portal; AI-DLC mob sessions; reduced compliance toil
Agents in production	23 (siloed)	120 (platform-governed)	Agent lifecycle management; health scoring; marketplace
MTTR for AI incidents	6.3 hours	< 30 minutes	LGTM observability; AI-DLC Operations; incident agent (Pioneer 4)

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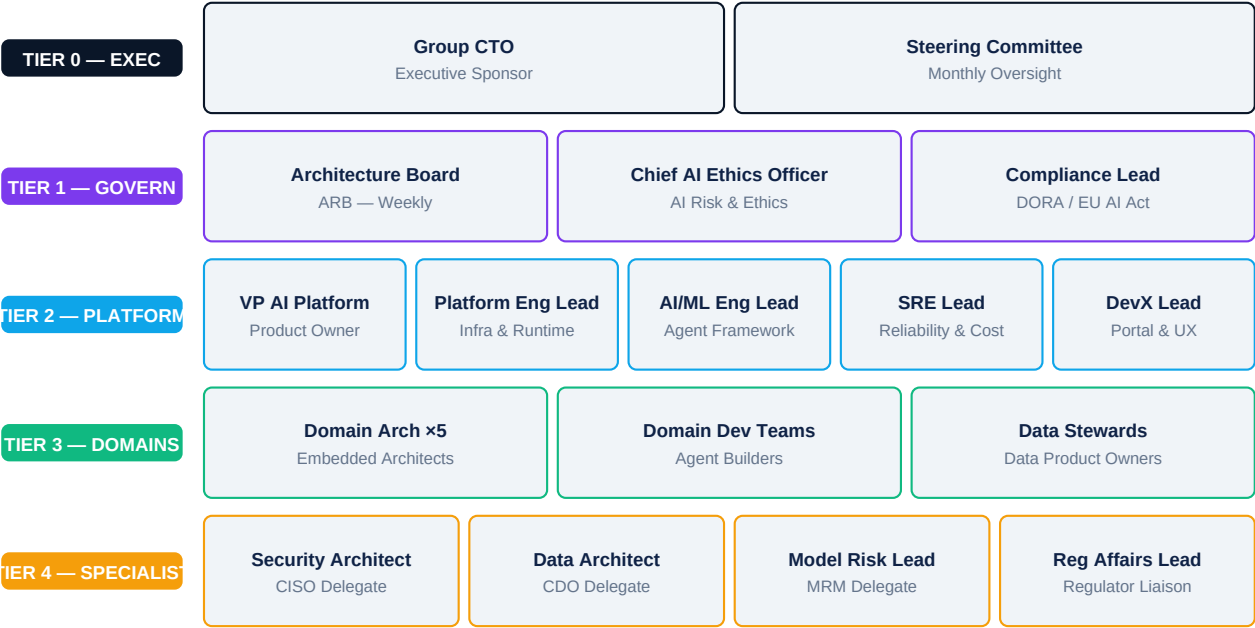
SECTION 01

Team Structure, RACI & Responsibility Mappings

Who is on the team, what they own, and how they interact

APEX Operating Model — Who Does What

The APEX programme operates a federated model: a **Platform Core Team** builds and runs the shared infrastructure; **Domain Squads** build and operate agents within it; **Governance Bodies** provide oversight and approval authority; and **Specialist Functions** (Security, Compliance, Data) provide embedded expertise throughout the lifecycle.



Core Team Register

Role ID	Role	Tier	FTE	Source	Key Accountabilities
T-01	Group CTO	0	0.1 (spon sor)	Internal	Executive authority; budget approval; regulator-facing sign-off
T-02	VP AI Platform	2	1.0	Internal	Platform product roadmap; stakeholder alignment; steering pack
T-03	Enterprise Architect	1	1.0	Internal	ADM governance; all architecture decisions; ARB chair
T-04	Chief AI Ethics Officer	1	1.0	New hire	AI risk appetite; ethics review; EU AI Act risk classification sign-off
T-05	Platform Engineering Lead	2	1.0	Internal	Cloud-native infrastructure; Kubernetes clusters; CI/CD platform
T-06	Platform Engineers (×4)	2	4.0	Internal (×2) + Contractor (×2)	IaC authoring; agent gateway; agent control plane build
T-07	AI/ML Engineering Lead	2	1.0	Internal	Agent framework selection; orchestration patterns; KB architecture
T-08	AI/ML Engineers (×3)	2	3.0	Internal (×1) + AWS PS (×2)	Agent development; prompt engineering; guardrail configuration
T-09	SRE / FinOps Lead	2	1.0	Internal	Platform reliability; cost governance; agent health scoring
T-10	SRE Engineers (×2)	2	2.0	Internal (×1) + Contractor (×1)	Observability; on-call; canary monitoring; budget enforcement
T-11	Developer Experience Lead	2	1.0	Internal	Self-service portal; developer documentation; agent marketplace UX

Role ID	Role	Tier	FTE	Source	Key Accountabilities
T-12	DevX Engineer	2	1.0	Internal	Portal frontend; onboarding wizard; developer tooling
T-13	Domain Architect (x5)	3	5.0	Internal — reassigned from domains	Domain-level architecture; Pioneer agent design; domain ARB liaison
T-14	Compliance Analyst (AI) (x2)	1	2.0	New hire	EU AI Act classification; DORA evidence; DEA review; audit liaison
T-15	Security Architect	4	0.5 (embedded)	Internal — CISO delegate	Security architecture sign-off; red-team coordination; zero-trust design
T-16	Data Architect	4	0.5 (embedded)	Internal — CDO delegate	Data classification; embedding governance; data mesh integration
T-17	Model Risk Lead	4	0.5 (embedded)	Internal — MRM delegate	Model risk opinions; explainability requirements; SR 11-7 alignment
T-18	Regulatory Affairs Lead	4	0.5 (embedded)	Internal — Group Compliance	Regulator liaison; DORA evidence pack; EU AI Act submissions

RACI Matrix — Agent Lifecycle

DOC REF: RACI-APEX-001

R = Responsible (does the work) A = Accountable (owns the outcome) C = Consulted (input required) I = Informed (kept up to date)

Activity	Ent. Arch	VP AI Plat	Platform Eng	AI/ML Eng	Domain Arch	Domain Dev	Security Arch	Data Arch	Model Risk	Compliance	SRE / FinOps
Agent onboarding request intake	C	A	I	I	R	C	I	I	I	C	I
Data Maturity Gate assessment	A	I	I	I	R	C	I	R	C	C	I
Agent architecture blueprint	A	C	C	C	R	C	R	R	C	C	I
Security design & threat model	C	I	C	I	C	I	R/A	C	I	I	I
Data classification & DPO sign-off	C	I	I	I	C	I	C	R/A	I	C	I
Prompt catalog design & review	I	I	C	R/A	C	R	I	C	I	C	I
Knowledge base architecture	C	I	C	R/A	C	R	I	R	I	I	I
IaC generation (AI-assisted)	I	I	R/A	C	C	R	C	I	I	I	I
L1–L5 verification pipeline	I	I	R/A	C	C	C	C	I	I	I	C
EU AI Act risk classification	C	I	I	I	C	I	I	I	C	R/A	I
Model risk validation opinion	C	I	I	C	C	I	I	I	R/A	C	I
DEA generation service config	C	I	R	R	C	C	C	I	C	A	I
Guardrail configuration	C	I	C	R/A	C	C	R	I	C	C	I
Canary deployment & SLO monitoring	I	I	C	I	I	C	I	I	I	I	R/A
Production go-live approval	A	C	C	I	C	I	C	C	C	C	C
Agent health score & budget review	I	C	C	I	C	I	I	I	I	I	R/A
Agent deprecation decision	C	A	C	I	R	C	I	I	I	I	C

RACI Matrix — Platform Governance & Architecture Board

Activity	Ent. Arch	VP AI Plat	Platform Eng	AI/ML Eng	Domain Arch	Domain Dev	Security Arch	Data Arch	Model Risk	Compliance	SRE / FinOps
ARB session facilitation	R/A	C	I	I	C	I	C	C	I	I	I
Architecture principle ratification	A	C	I	I	C	I	C	C	C	C	I
Architecture deviation approval	A	C	I	I	C	I	C	C	I	C	I
New SBB build/buy decision	A	R	C	C	C	I	C	C	I	C	C
Architecture debt prioritisation	R/A	C	C	C	C	I	I	I	I	I	C
Conformance report (T1 automated)	I	I	R/A	I	I	I	C	I	I	I	C
Conformance escalation (T2 arch)	R/A	C	C	I	C	I	C	I	I	I	I
T4 Regulatory Gate sign-off	C	I	I	I	I	I	C	C	C	R/A	I
Executive steering pack	C	R/A	I	I	I	I	I	I	I	C	I
Agent marketplace curation	C	A	C	I	C	R	I	I	I	I	I
Platform SLA reporting	I	C	C	I	I	I	I	I	I	I	R/A
FinOps & chargeback reporting	I	A	R	I	I	I	I	I	I	I	R

RACI Matrix — Regulatory & Compliance

Activity	Ent. Arch	VP AI Plat	Platform Eng	AI/ML Eng	Domain Arch	Domain Dev	Security Arch	Data Arch	Model Risk	Compliance	SRE / FinOps
EU AI Act risk tier classification	C	I	I	I	C	I	I	I	C	R/A	I
DEA completeness review	I	I	C	I	I	I	C	I	C	R/A	I
DORA change record creation	I	I	R	I	C	I	C	I	I	A	I
DORA 72-hr incident notification	C	C	C	I	I	I	C	I	I	R/A	C
DORA evidence pack compilation	C	I	C	I	C	I	C	C	C	R/A	I
GDPR data flow map maintenance	C	I	I	I	C	I	C	R/A	I	C	I
Model risk opinion sign-off	C	I	I	C	C	I	I	I	R/A	C	I
Explainability evidence (SHAP/LIME)	I	I	C	R	C	C	I	I	A	C	I
Regulator liaison & submissions	C	C	I	I	I	I	I	I	C	R/A	I
TLPT (penetration testing) oversight	C	I	C	I	I	I	R/A	I	I	C	C
Compliance dashboard maintenance	I	I	R	I	I	I	I	I	I	A	I

SECTION 01 cont.

Responsibility Assignment Matrix & Stakeholder Map

Domain ownership, stakeholder power/interest, and engagement model

Responsibility Assignment Matrix — Platform Capabilities

The RAM maps each APEX platform capability to its accountable owner and key contributors. This extends the RACI to show structural ownership across the platform.

Platform Capability	Accountable Owner	Primary Team	Secondary Contributors	Governance Body	Review Cadence
Agent Lifecycle Management (ALM)	VP AI Platform	AI/ML Engineering	Domain Architects, Compliance, Model Risk	ARB	Per deployment
Agent Gateway & Registry	Platform Engineering Lead	Platform Engineers	Security Architect, SRE	ARB (new agents)	Continuous (automated)
Knowledge Base Platform	AI/ML Engineering Lead	AI/ML Engineers	Data Architect, Domain Data Stewards	ARB + CDO delegate	Per KB refresh
Guardrails & Safety Layer	Chief AI Ethics Officer	AI/ML Engineering	Security Architect, Compliance	ARB + Ethics review	Per agent + quarterly red-team
Decision Explanation Artefact (DEA)	Compliance Lead	Platform Engineering	Model Risk, Legal, AI/ML Engineering	T4 Reg-Gate	Per regulated decision
CI/CD Pipeline & Layered Verification	Platform Engineering Lead	Platform Engineers	Security Architect, SRE	ARB (pipeline changes)	Per pipeline change
Observability & LGTM Stack	SRE Lead	SRE Engineers	Platform Engineering	SRE review	Continuous + weekly review
Cost Governance (FinOps)	SRE Lead	SRE Engineers	VP AI Platform, Finance	Monthly steering	Monthly
Developer Portal & Marketplace	DevX Lead	DevX Engineer	VP AI Platform, Domain Architects	Quarterly UX review	Per release
Regulatory Compliance Programme	Compliance Lead	Compliance Analysts	All Tier 4 specialists	Steering Committee	Monthly + ad-hoc
Data Mesh Integration	Data Architect	Domain Data Stewards	AI/ML Engineering, Platform Engineering	CDO governance forum	Per data product change
Security Architecture	Security Architect	Platform Engineering	All teams	CISO review board	Per phase + quarterly
Pioneer Domain 1 — Risk Scoring Agent	Domain Architect 1	Domain Dev Team 1	AI/ML Engineering, Model Risk, Compliance	Domain ARB	Per sprint
Pioneer Domain 2 — Verification Agent	Domain Architect 2	Domain Dev Team 2	AI/ML Engineering, Compliance	Domain ARB	Per sprint
Pioneer Domain 3 — Advisory Agent	Domain Architect 3	Domain Dev Team 3	AI/ML Engineering, Model Risk	Domain ARB	Per sprint
Pioneer Domain 4 — Incident Agent	Domain Architect 4	Domain Dev Team 4	SRE, Platform Engineering	Domain ARB	Per sprint
Pioneer Domain 5 — Model Monitor Agent	Domain Architect 5	Domain Dev Team 5	Model Risk, Compliance, AI/ML Eng	Domain ARB + MRM	Per sprint

Stakeholder Power / Interest Map

Stakeholder engagement strategy is derived from power (organisational authority and decision-making ability) and interest (level of direct impact from APEX).

Stakeholder	Role	Power	Interest	Quadrant	Strategy	Engagement Cadence
Group CTO	Executive Sponsor	High	High	KEY PLAYER	Manage closely; regular updates; co-own critical decisions	Monthly steering; ad-hoc for critical decisions
VP AI Platform	Programme Owner	High	High	KEY PLAYER	Daily collaboration; roadmap co-ownership; ARB participation	Daily standup; weekly roadmap review
Chief Information Security Officer	Security Authority	High	High	KEY PLAYER	Embedded security architect; CISO sign-off gate in ADM	Bi-weekly architecture review; CISO gate per phase
Group Compliance Lead	Regulatory Liaison	High	High	KEY PLAYER	Co-own compliance gates; evidence pack review; regulator briefings	Monthly evidence review; weekly during regulatory events
Chief Data Officer	Data Governance Authority	High	High	KEY PLAYER	Embedded data architect; Data Mesh integration authority	Bi-weekly; data architecture sign-off
Chief AI Ethics Officer	AI Risk Authority	High	High	KEY PLAYER	Direct line into ARB; EU AI Act classification authority	ARB weekly; ethics review per agent class
Chief Risk Officer	Enterprise Risk Authority	High	Medium	MANAGE CLOSELY	Quarterly risk briefings; model risk programme alignment	Quarterly + escalation for material AI risks
Chief Financial Officer	Budget Authority	High	Medium	MANAGE CLOSELY	Monthly FinOps reports; business case sign-off	Monthly FinOps review; annual budget cycle
Industry Regulator (PRA/ECB)	External Oversight	High	Medium	MANAGE CLOSELY	DORA evidence pack; EU AI Act submissions; proactive briefings	Quarterly regulatory briefing
Domain COOs (x5)	Pioneer Domain Sponsors	Medium	High	KEEP INFORMED	Regular progress updates; Pioneer milestone reviews	Monthly per domain
Domain Dev Teams	Agent Builders	Low	High	KEEP INFORMED	Self-service tooling; developer portal; clear onboarding	Bi-weekly sprint reviews; open ARB office hours
Internal Audit	Assurance Function	Medium	Medium	KEEP INFORMED	Annual audit; DORA evidence access; architecture artefacts	Annual audit; ad-hoc evidence requests
HR / People Team	Change Management	Low	Medium	MONITOR	Change impact comms; role transition support for displaced tools	Quarterly change update
External Cloud Providers	Technology Partners	Low	Low	MONITOR	SLA management; roadmap briefings	Quarterly account review

Team Interaction & Dependency Map

The following matrix shows how each team tier interacts with others, the nature of the dependency, and the formal channel for that interaction.

From → To	Platform Core Team	Domain Squads	ARB / Governance	Specialist Functions	External Stakeholders
Platform Core Team	Internal: sprint reviews, platform roadmap sync	Provides: infrastructure, tooling, self-service portal, CI/CD pipeline, guardrails	Reports to: ARB weekly; steering monthly. Seeks: approval for SBBs and deviations	Consults: Security (each phase), Data (KB design), Model Risk (agent risk)	Cloud provider SLA management; AWS PS engagement for AI/ML

From → To	Platform Core Team	Domain Squads	ARB / Governance	Specialist Functions	External Stakeholders
Domain Squads	Consumes: platform services, agent templates, portal, pipeline. Raises: new agent requests, escalations	Internal: daily standups, shared domain backlog, cross-domain agent dependency reviews	Reports to: Domain ARB. Seeks: architecture approval, compliance sign-off per agent	Consults: Domain Architect (daily), Compliance (per agent intake), Data Steward (data flows)	Business stakeholder demos; UAT with end users
ARB / Governance	Approves: architecture decisions, SBBs, deviations, phase gate exits	Reviews: agent blueprints, domain architecture artefacts, compliance evidence	Internal: ARB weekly chair + secretariat. Escalates: to Steering Committee for T5 items	Receives: security review (CISO gate), compliance opinion (T4 Reg-Gate), Model Risk opinion	Regulator liaison via Compliance Lead; quarterly briefings
Specialist Functions	Embedded in: security design, data architecture, CI/CD gates	Provides: domain-level guidance, data classification, embedded model risk reviews	Feeds into: ARB with specialist opinions. Attends: relevant ARB agenda items	Cross-functional: security–compliance alignment; data–model risk alignment	No direct external engagement; all via Compliance Lead or CISO
External Stakeholders	Receives: SLA reports, FinOps dashboards, compliance packs	None (no direct domain team interaction)	Attends: quarterly regulator briefing. Reviews: DORA evidence pack, EU AI Act submissions	None direct	Peer industry: AI governance working groups; open-source community participation

SECTION 02

AI-DLC: Methodology, Phases & EA Impacts

AI as central collaborator — not peripheral tool

What is AI-DLC?

The **AI-Driven Development Lifecycle (AI-DLC)** was introduced by AWS in July 2025 and featured at AWS re:Invent 2025. It emerged from 100+ enterprise experiments and observed that AI-assisted (too narrow) and AI-autonomous (too unreliable) both produced suboptimal results. AI-DLC positions AI as a **central collaborator** throughout every development activity — design, build, test, deploy, and operate.

Proven Results

Enterprise adopters report 2–5× sustainable productivity gains across the full lifecycle. In well-scoped greenfield tasks with high-quality semantic context, gains reach 7–10×. Teams using AI-DLC consistently report improved code quality alongside velocity: the Qodo 2025 report showed quality improvements rising from 55% to 81% with AI-assisted review; the Atlassian RovoDev 2026 study found AI code review comments led to additional fixes in 38.7% of cases.

Phase	Name	AI Role	Human Role	Architecture Impact
1	INCEPTION (Mob Elaboration)	Transforms business intent into structured requirements, user stories, capability maps, and architecture units at speed	Validates AI outputs; provides business context, domain knowledge, and political nuance that AI cannot infer	Vision, requirements, and capability maps produced in days not weeks; must be preceded by a political context capture session
2	CONSTRUCTION (Mob Construction)	Proposes architecture components, generates IaC, writes serverless functions, builds tests from semantic context	Clarifies technical decisions and architectural choices in real time; reviews every AI-generated artefact before merge	Phases B/C/D produced concurrently in bolts; concurrent execution requires Phase Boundary Receipts for regulated data flows
3	OPERATIONS	Applies accumulated context to IaC deployment, canary monitoring, incident pattern recognition, and drift detection	Reviews and approves all changes; monitors SLOs; makes judgment calls in novel failure scenarios	Phases F/G/H compressed into continuous delivery; every AI-driven change is a formal DORA change event with ARB record

Terminology Reference — AI-DLC vs. Traditional

Traditional Term	AI-DLC Term	Duration Change	Enterprise Architecture Implication
Sprint	Bolt	2–4 weeks → hours to days	ADM governance must operate within bolt cadence; tiered model required
Epic	Unit of Work	Months → days to weeks	Architecture work packages shrink; ARB meets weekly not bi-weekly
User Story	AI-Elaborated Requirement	Days → minutes	Requirements traceability automated; ARS auto-updated with human review gate
Architecture Review	Real-time Mob Construction	Weeks → concurrent	ARB provides real-time guidance; final sign-off authority retained
Documentation	Persistent Semantic Context	Post-sprint → continuous	Architecture repository auto-updated; EA team shifts from authors to reviewers
Code Review (PR)	AI-Augmented Quality Scan	Hours → minutes	L1–L5 Layered Verification Model enforced in pipeline; no hero review
Retrospective	Continuous Learning Loop	End of sprint → continuous	Architecture debt auto-detected; agent health scoring continuous

AI-DLC Impact Across All 10 TOGAF ADM Phases

TOGAF Phase	AI-DLC Capability	Human Guard Required	Architecture Adaptation
Preliminary	AI generates architecture principle drafts; governance rules codified as steering files	Architects validate all AI-generated principles; formal ratification gate mandatory	Steering files maintained in version control; breaking changes require architect approval
Phase A — Vision	AI transforms business intent into draft Architecture Vision and SAW in hours	Political nuance, power dynamics, and unwritten constraints require human capture before Mob Elaboration	Political Blindspot Protocol precedes every Mob Elaboration session
Phase B — Business	AI auto-generates business capability maps from existing documentation; value streams modelled in minutes	Existing documents may be stale; Data Maturity Gate required before AI elaboration begins	Data Maturity Gate: 5-check assessment mandatory before Phase B AI execution
Phase C — Data	AI creates data models, entity diagrams, and API schemas; prompt catalogs and embeddings become first-class governed entities	Embedding model upgrades require Embedding Compatibility Contract and RAGAS regression gate	Embeddings versioned; RAGAS gate on model upgrade; 90-day rollback window maintained
Phase C — Application	AI proposes application component models and cloud-native integration patterns	AI may propose patterns inconsistent with org standards without tight steering files	Org standards loaded as steering files; all AI-proposed patterns reviewed by ARB
Phase D — Technology	AI generates IaC, security policies, and deployment configs from architecture context	AI-generated IaC requires L1–L5 Layered Verification; IaC hallucinations are a real risk	L1–L5 Layered Verification Model enforced in CI/CD; no merge without engineer confirmation
Phase E — Opportunities	AI analyses gap reports and proposes build/buy/reuse options at machine speed	AI may miss regulatory obligations, vendor relationship context, and political constraints	Legal, procurement, and compliance participate in Mob Elaboration for all build/buy decisions
Phase F — Migration	Bolts replace work packages; migration plans near-real-time	Over-acceleration may skip governance gates	Compliance gates pipeline-enforced; AI-DLC speed contained within governance boundaries
Phase G — Governance	AI automates conformance checking; real-time drift detection replaces periodic reviews	Automated checks inform; humans decide; ARB retains final sign-off authority	5-tier governance model; T1 automated always-on; T4 Reg-Gate inviolable
Phase H — Change	AI monitors model releases and regulatory changes; flags architecture impacts automatically	Continuous change proposals require human filtering to avoid architecture instability	Change type classification; AI flags; humans prioritise; DORA change framing applied to all AI changes
Req. Management	AI generates and maintains requirements traceability from business intent to deployed code	Context drift across sessions; requirements must be forward-traced before each deployment	Context Integrity Protocol; session summary archive; forward-trace validation per deployment

PRELIMINARY

Preliminary Phase — Foundation

Architecture capability, principles, tailored ADM, repository

The Preliminary Phase establishes GlobalCorp's architecture capability for APEX. Governance structures are put in place, the ADM is tailored for regulatory context, and architecture principles that govern every subsequent decision are ratified.

Architecture Principles — APD-001

DOC REF: APD-001 | APPROVED

Business Principles

ID	Principle	Statement	Implication
BP-01	AI as Business Capability	Every APEX agent traces to a measurable business outcome in the Corporate Strategy Map; productivity gains are tracked quarterly against actuals	Business Value Case required before APEX onboarding; quarterly actuals vs. targets reported to steering committee
BP-02	Human-in-the-Loop — Inviolable	All agents acting on sensitive data or high-stakes decisions have a human escalation path that is pipeline-enforced and cannot be bypassed for velocity	HITL gates configured in the agent gateway; any removal requires Group CTO written exception logged in compliance register
BP-03	Platform Thinking with Sprawl Controls	No division procures standalone AI tools if APEX can meet the need within 90 days; equally, no agent is created without registry entry and monthly health score	ARB approval for procurement; agent registry is the source of truth; unregistered agents blocked at gateway
BP-04	Data Maturity Before AI Velocity	AI-DLC Construction cannot begin for any domain until the Data Maturity Gate (5 checks) is passed	Data maturity assessment is the first deliverable of every Pioneer Domain onboarding; it is a programme gate not a guideline

Data Principles

ID	Principle	Statement	Implication
DP-01	Data Sovereignty	Personal and sensitive data processed by APEX agents must remain within the contractually agreed jurisdiction	AI inference endpoints provisioned per jurisdiction; network policy blocks cross-border calls for C4-class data at the kernel level
DP-02	Data as Shared Asset	All data consumed by agents is catalogued in the Data Mesh with documented lineage before an agent goes to production	No agent deployed without data catalog entry covering every data source it touches; Data Architect sign-off required
DP-03	Explainability Before Deployment	Every model has a documented explainability method — including the retrieval stage of RAG, not just the generation stage	RAG retrieval explanation is part of the DEA; explainability report is a mandatory CI/CD gate artefact
DP-04	AI Assets as Governed Data	Prompt templates, embeddings, feedback logs, and model lineage records are first-class governed data assets with the same rigor as operational data	Embedding Compatibility Contract applies; ML model registry tracks all lineage; DPO sign-off for sensitive-adjacent prompts
DP-05	Embedding Compatibility Enforced	Any embedding model upgrade triggers automatic full re-indexing of all dependent knowledge bases; RAGAS regression gate blocks production activation	CI/CD pipeline blocks embedding upgrade without RAGAS pass; 90-day rollback window maintained

Technology Principles

ID	Principle	Statement	Implication
TP-01	Cloud-Native Open Standards	Platform uses CNCF-hosted and open-standard components; proprietary cloud SDKs used only where equivalent OSS is not production-ready	All components containerised; OpenAPI for agent interfaces; OpenTelemetry for observability; no cloud SDK in core business logic
TP-02	Security by Design	Security architecture is approved by the CISO delegate before any IaC is written for a given Pioneer Domain	Security is a Phase D entry gate; it is not a Phase G concern; zero-trust NetworkPolicy defined before build begins
TP-03	Observability as First-Class	Every agent emits OpenTelemetry traces, Prometheus metrics, and structured logs from the first commit; no agent reaches UAT without a Grafana dashboard	Dashboard-as-code is generated in bolt 1; observability is part of the Definition of Done for every agent
TP-04	Cost Transparency with Hard Throttle	Each agent has a tagged cost centre; budget enforced with a hard throttle at 110% of monthly budget that cannot be self-overridden by the agent	Budget throttle is enforced at the agent gateway level, not just an alert; SRE Lead is the only person who can override with a logged justification
TP-05	Layered Verification	AI-generated IaC passes five verification layers (L1 static analysis, L2 AI explanation artefact, L3 property tests, L4 canary deploy, L5 drift detection) before merge; no single-person hero review is sufficient at enterprise scale	CI/CD enforces L1–L4 automatically; L2 AI explanation artefact retained 12 months per IaC commit
TP-06	Decision Explanation Artefact by Default	Every High-Risk AI agent (EU AI Act Art.6) produces a Decision Explanation Artefact (DEA) for every regulated decision — assembled from all log sources and stored immutably for 7 years	DEA generation service is a mandatory platform component; deployed with Pioneer 1 and applied to all subsequent Pioneers

PHASE A

Architecture Vision

Scope, mandate, and stakeholder alignment

Architecture Vision Statement

"By Q4 2026, GlobalCorp will operate a unified AI Agent Platform (APEX) on cloud-native open standards, reducing time-to-market for AI use cases to a planning baseline of 5–6 weeks, achieving full EU AI Act and DORA compliance, and generating positive NPV within 24 months through platform economics. Productivity gains are tracked quarterly against actuals."

Programme Scope

In scope: Cloud-native multi-region agent platform (4 regions); AI agent lifecycle management; multi-agent orchestration; self-service developer portal; shared knowledge bases; centralised observability and cost management; EU AI Act compliance automation; DEA generation; Agent Gateway; and 5 Pioneer Domain agents.

Out of scope: Foundation model training; customer-facing AI chatbots (Digital Channel programme); autonomous execution agents (requires separate regulatory approval pathway); data mesh platform build (existing programme).

Architecture Constraints — SAW-APEX-001

DOC REF: SAW-APEX-001 | APPROVED – Group CTO sign-off

Constraint	Source	Architecture Impact
Personal and sensitive data must remain within contractual jurisdiction	GDPR / PDPA / LGPD	Multi-region deployment; network policy blocks cross-border agent calls for personal data at kernel level
All AI models require Model Risk validation before production	Internal Model Risk Policy (aligned to SR 11-7)	Model Risk Gate in CI/CD is a T4 Reg-Gate; DEA required for High-Risk agents; explainability artefact mandatory
Cloud-native open standards; no proprietary SDK in core business logic	TP-01	LLM calls via provider-agnostic adapter library; agent logic portable across cloud providers; OpenAPI for all agent interfaces
Zero-downtime deployment for critical services	Enterprise SLA	Argo Rollouts canary; SLO-based auto-rollback; no manual blue-green steps required
EU AI Act enforcement from August 2026	EU Regulation 2024/1689	High-Risk classification at intake; DEA pattern deployed from Pioneer 1; compliance pack (WP-012) delivered in H2
All AI evolution must be framed as controlled DORA change	DORA Art.11	Every model update, KB refresh, and agent topology change is a formal change event with ARB record and DORA change log

Five Pioneer Domains — Proof of Value

#	Division	Agent	Use Case	Key Regulation	AI-DLC Bolts	Owner
1	Customer Division	RiskScoringAgent	Automated risk-based decisioning with human escalation for high-value cases	Internal Model Risk Policy, GDPR	3 bolts (~8 weeks)	Domain Architect 1
2	Enterprise Division	VerificationAgent	Continuous customer verification monitoring, sanctions screening, and report drafting	AML Directive, FATF 40	3 bolts (~7 weeks)	Domain Architect 2
3	Advisory Division	RebalancingAgent	Rule-based portfolio rebalancing recommendations with full suitability assessment	MiFID II Art.27, GDPR	2 bolts (~6 weeks)	Domain Architect 3

#	Division	Agent	Use Case	Key Regulation	AI-DLC Bolts	Owner
4	IT Operations	IncidentAgent	L1/L2 ticket triage, root-cause pattern recognition, and auto-remediation with SRE oversight	DORA Art.11 ICT Risk	2 bolts (~5 weeks)	Domain Architect 4
5	Risk & Compliance	ModelMonitorAgent	Model drift detection, backtesting orchestration, and regulatory evidence report generation	EU AI Act Art.9, Internal Policy	3 bolts (~6 weeks)	Domain Architect 5

PHASE B

Business Architecture

Capability model, value streams, and adaptive capability governance

Business Capability Map — BCM-APEX-001

DOC REF: BCM-APEX-001 | APPROVED

L1 Capability	Type	Critical Gaps	APEX	AI-DLC Adaptive?	Governance Model
1. AI Strategy & Governance	STATIC	4 critical	5/5	Partial — policy enforcement via agents	Standard ARB quarterly review
2. Agent Development	ADAPTIVE	5 critical	5/5	YES — AI-DLC Construction	Bolt-level ARB; L1–L5 verification; bolt Definition of Done
3. Agent Orchestration	ADAPTIVE	4 critical	4/4	YES — AI-DLC Operations	Agent dependency map maintained; topology change = formal DORA change event
4. Knowledge Management	ADAPTIVE	4 critical	4/4	YES — continuous KB refresh	KB refresh = content change event; RAGAS gate; DPO sign-off for sensitive content
5. AI Operations (AIOps)	ADAPTIVE	5 critical	5/5	YES — AI-DLC Operations	DORA change framing applied; real-time monitoring; agent health scoring continuous
6. Data Integration	STATIC	2 critical	3/4	PARTIAL — Data Mesh governed	Data Mesh domain governance; quarterly data product review; APEX integrates, not owns
7. Platform Enablement	STATIC	4 critical	4/4	YES — portal AI-scaffolded	Standard deployment cadence; portal releases are standard CI/CD pipeline deployments

Value Stream — Agent Delivery End-to-End

Stage	Baseline	APEX Plan	AI-DLC Planning Baseline	Key Constraints
Intake & Qualification	2–4 weeks	5 days	2 days	Data Maturity Gate may add 1–2 weeks in early cohorts (F-06)
Architecture & Design	4–8 weeks	10 days	4–5 days	ARB Gate cannot be bypassed; political context capture session adds structured time in complex domains
Build & Integrate	8–16 weeks	3 weeks	2 weeks	L1–L5 verification non-negotiable; DEA service adds 1 bolt for High-Risk agents
Validate & Govern	6–10 weeks	10 days	7–8 days	T4 Reg-Gate (Model Risk opinion) is inviolable; explainability artefact mandatory for High-Risk
Deploy & Monitor	2–4 weeks	1 week	3 days	DORA change framing adds a compliance record step; canary monitoring adds 1–2 days
TOTAL TIME-TO-MARKET	~9 months	~9.5 weeks	5–6 weeks	Planning baseline used for budgets; 9.5 weeks used for business cases as conservative estimate

PHASE C

Information Systems Architecture

Data, AI-native entities, application components, and API catalogue

Data Classification Framework

DOC REF: DAD-APEX-001 | APPROVED

Class	Description	Examples	Guardrail Action	Retention
C4 — Highly Restricted	PII and highly sensitive data	Government IDs, account numbers, biometrics	BLOCK or ANONYMIZE in all agent logs and traces	7 years (DORA mandate)
C3 — Confidential	Sensitive internal business data	Proprietary models, strategic risk parameters	Redact in all observability traces and log exports	5 years
C2 — Internal	Non-sensitive operational data	Agent performance metrics, usage statistics	Standard structured logging	3 years
C1 — Public	Regulatory and market reference data	EU AI Act text, public rates, open standards	No restriction	Indefinite
C-AI — AI Assets	Prompt catalogs, embeddings, feedback logs, model lineage	Versioned prompt templates, vector embeddings, RLHF datasets	Inherit from source data class; version-controlled; DPO sign-off for C3/C4-adjacent	Per source data class rules

AI-Native Data Entities — Governance Model

Entity	Classification	Governance Owner	Version Control	Key Control	Retention
Prompt Catalog	C-AI	Platform AI Eng + DPO	Git semver; PR mandatory	Prompt change triggers full regression test suite before merge	Per source data class
Embedding Store	C-AI	Data Mesh Domain Owner	Embedding model ID + version in data catalog metadata	Embedding Compatibility Contract: RAGAS gate on model upgrade; 90-day rollback	90 days post-supersession
RAG Retrieval Log	C2→C4 inherit	Compliance + Audit	Immutable; object lock; append-only	Queryable by compliance team within 60 seconds; feeds DEA assembly	7 years (DORA)
Feedback Log	C2→C4 inherit	Model Risk Management	Immutable; object lock; append-only	Drives model drift detection; feeds Model Monitoring Agent (Pioneer 5)	7 years (DORA)
Agent Memory	Inherits source class	Platform Eng + DPO	TTL enforced in KV store; max 30 minutes	PII check on every write; TTL enforced at gateway; no persistence after session	Session TTL only
Model Lineage Record	C3	Model Risk Management	ML model registry (MLflow equivalent)	New model version = formal model update event requiring ARB record	Model lifecycle + 7 years
Decision Explanation Artefact	C3	Compliance + Legal	Generated at decision time; immutable; DEA service validates completeness	Assembled from all log sources; queryable on demand; 7-year immutable retention	7 years (EU AI Act + DORA)

Application Architecture — Cloud-Native Component Model

DOC REF: AAD-APEX-001 | APPROVED

Component	Technology	Pattern	SLA	Key Design Decision
APEX Developer Portal	React SPA + API Gateway (Kong/Nginx/cloud-native)	Backend-for-Frontend (BFF)	99.9% / p99 <500ms	Agent health score and DEA query link visible per agent; AI-DLC scaffolds UI in 1 bolt
APEX Control Plane	Python FastAPI; Kubernetes OCI containers; 6 microservices	Microservices — bounded contexts	99.9% / p99 <500ms	Agent Lifecycle, Budget Enforcer, Onboarding Orchestrator, Chargeback, Notification, Compliance Reporter services
Agent Gateway	Envoy proxy + custom plugin OR APISIX + plugin	Zero-trust entry point; registry lookup; budget throttle	99.99% / p99 <50ms	Blocks unregistered agents (403); enforces budget hard-throttle at 110%; logs all agent dependency calls
Agent Orchestration Engine	LangGraph / AutoGen / CrewAI OR cloud-managed equivalent (evaluated in WP-002 spike)	Supervisor-Worker multi-agent pattern	99.95% / p99 <3s	All routing decisions logged to OpenTelemetry trace for DEA assembly; abstraction layer prevents framework lock-in
Serverless Tool Functions	Knative / OpenFaaS / cloud FaaS (OCI container-based)	Action Group / Tool Sidecar pattern	99.95% / p99 <1s	Each function passes L1–L5 verification; tool invocation logs written to structured log for DEA
DEA Generation Service	Python event-driven service; Kubernetes; triggered on regulated decision	Event-driven async; completeness validation before store	99.9% / <5s	Assembles DEA from all log sources; validates completeness; stores immutably; queryable via DEA API
Human Review Portal	React + Temporal / Camunda workflow engine	Human-in-the-Loop (HITL) review workflow	99.5% / SLA defined per agent risk tier	HITL outcome written to DEA; DEA marked complete only after HITL where required by risk classification
LGTM Observability Stack	OpenTelemetry Collector + Prometheus + Grafana + Loki + Tempo	Centralised OTEL collector; push-based metrics and traces	99.5% ingestion; real-time dashboards	Agent health score metrics; emergent behaviour alert rules (chain >3 hops or latency >10s)
Compliance Reporter	Python data pipeline + Apache Superset or Metabase	Batch + event-driven reporting	99.5% daily run	EU AI Act and DORA report templates; DORA change record auto-writer; DEA export API

API Catalogue

API	Version	Style	Auth	SLA	Owner
Agent Registry API	v2	REST / OpenAPI 3.1	OAuth 2.0 (APEX scope)	99.9% / p99 <200ms	Platform Eng
Agent Invocation API	v1	REST + SSE streaming	IAM / mTLS	99.95% / p99 <3s	Platform Eng
Knowledge Base API	v1	REST / JSON	IAM / mTLS	99.9% / p99 <500ms	AI/ML Eng
Onboarding API	v1	REST / JSON	OAuth 2.0	99.5% / p99 <1s	Platform Eng
DEA Query API	v1	REST / JSON	OAuth 2.0 + MFA	99.9% / <60s response	Compliance
Chargeback API	v1	REST / JSON	mTLS	99.9% monthly run	SRE / FinOps
Compliance Report API	v1	REST / JSON	OAuth 2.0 + MFA	99.5% daily run	Compliance

PHASE D

Technology Architecture

Cloud-native infrastructure, security, CI/CD with Layered Verification

DOC REF: TAD-APEX-001 | APPROVED

Multi-Region Deployment Model

Region	Role	Data Scope	Key Platform Services	Regulatory Basis
Region 1 — Primary (EU/UK)	ACTIVE PRIMARY	EU/UK personal + sensitive data	Kubernetes cluster (multi-AZ); LGTM stack; agent gateway; vector store; PostgreSQL multi-AZ; Kafka; Vault; object storage (WORM)	GDPR, UK GDPR, DORA
Region 2 — DR (EU)	STANDBY / FAILOVER	DR replica only; no new data processing	Kubernetes cluster (standby); object storage (CRR target); DB replica	DORA Art.11 ICT resilience; GDPR Art.32
Region 3 — Americas	ACTIVE	US/LATAM institutional + customer data	Mirror of Region 1 topology; independent control plane and agent gateway	CCPA, LGPD, FINRA (where applicable)
Region 4 — APAC	ACTIVE	SG/HK/AU customer data	Mirror of Region 1 topology; regional vector store; local object storage	PDPA (SG), PIPL (CN), Privacy Act (AU)

Security Architecture

Security Domain	Control	Cloud-Native Implementation	Compliance Basis
Identity & Access	Zero-trust; OIDC federation; short-lived credentials for all service accounts	Service mesh mTLS (Istio/Linkerd); Kubernetes RBAC; OIDC-based identity federation	NIST SP 800-207, DORA Art.9
Encryption at Rest	AES-256; customer-managed keys; separate CMK per data classification	HashiCorp Vault or cloud KMS; 365-day automatic key rotation enforced	GDPR Art.32, ISO 27001
Encryption in Transit	TLS 1.3 minimum; mTLS between all internal services	Istio/Linkerd service mesh enforces mTLS; ingress enforces TLS 1.3 minimum externally	FIPS 140-2, DORA
Network Isolation	Zero-trust NetworkPolicy; no public internet egress from agent runtime pods	Kubernetes NetworkPolicy; egress via controlled proxy; PrivateLink-equivalent for cloud API calls	TP-01, Security Policy
Agent Safety	Content guardrail layer; grounding threshold 0.75; PII anonymisation	NeMo Guardrails or Guardrails AI; integrated in agent invocation path before output to caller	EU AI Act Art.9, BP-02
IaC Security	L1 static analysis on every commit; no merge on any HIGH or CRITICAL finding	Checkov + Semgrep + Trivy in CI; Falco at runtime; OWASP Dependency Check	CIS Kubernetes, DORA
Prompt Injection	Input sanitisation and validation on all agent inputs before model invocation	Validation middleware in agent gateway; OWASP LLM Top 10 controls; red-team testing quarterly	OWASP LLM Top 10
Secrets Management	All secrets in secrets manager; zero hardcoded credentials; 90-day rotation	HashiCorp Vault or cloud-native secrets manager; Checkov enforces in CI	CIS Kubernetes, DORA

CI/CD Pipeline — Layered Verification Model (L1–L5)

Stage	Tools	AI-DLC Role	Gate — Bypassable?
L1 Static Analysis	Checkov, Semgrep, Trivy, KICS, yamllint	AI explains failures and proposes fixes; cannot self-approve	NEVER — pipeline blocks merge; no override path exists
L2 AI Explanation Artefact	AI coding assistant; explanation stored as PR artefact (12 months)	AI explains every generated block before engineer approval; explanation is retained	NEVER — engineer confirms AI explanation before approving
L3 Property Tests	Terratest, pytest-infra, contract tests	AI generates test bodies; engineer writes property assertions that define correctness	NEVER — 100% pass required
RAGAS Evaluation Gate	RAGAS eval suite; custom GlobalCorp eval harness	AI generates adversarial test cases; regression scored against baseline	NEVER — >90% score required; regression blocks merge
Guardrail Test Suite	NeMo Guardrails test harness; adversarial prompt library	AI generates edge-case and adversarial prompts to test guardrail coverage	NEVER — 100% pass required
T4 Model Risk Gate	Ticket-based approval workflow (Jira / Linear / equivalent)	AI generates Model Risk evidence pack draft; Model Risk Lead reviews and signs	NEVER — human sign-off mandatory; T4 Reg-Gate; no automation path
L4 Canary Deploy	Argo Rollouts / Flagger; Prometheus SLO monitoring	AI monitors SLOs in real time; triggers automatic rollback on breach	SRE Lead override only; every override logged with justification
DEA Validation	DEA generation service completeness check; validation before production mark	DEA assembled from all log sources; completeness validated	NEVER for High-Risk AI agents
Post-Deploy DORA Record	Argo CD drift detection; cost tag validator; DORA change record writer	AI generates stakeholder summary; DORA change record auto-created with ARB reference	NEVER — records are mandatory; absence triggers T4 alert

PHASES E–H

Opportunities, Migration, Governance & Change

Roadmap, SBBs, 5-tier governance, and change intelligence

Architecture Roadmap — ROAD-APEX-001

DOC REF: ROAD-APEX-001 | APPROVED

Work Package	WP	Bolts	Wks	Horizon	Risk Level
Multi-region Kubernetes, zero-trust IAM, HashiCorp Vault, network policy	WP-001	2	4	H1	Low — well-understood pattern; IaC-automated
AI Agent Orchestration platform: framework spike, guardrails, KB-001	WP-002	2	4	H1	Medium — framework lock-in risk; abstraction layer required
CI/CD L1–L5 verification + DEA generation service deployment	WP-003	2	3	H1	Medium — DEA service is new pattern; allow spike time in bolt 1
Security baseline: zero-trust, mTLS, guardrails layer, Agent Gateway	WP-004	2	4	H1	Low — automated; well-understood pattern
LGTM observability stack + agent health score metrics + alerting	WP-005	1	2	H1	Low
Pioneer 1 — Risk Scoring Agent (first full end-to-end; all gates)	WP-006	3	8	H1	HIGH — first end-to-end run; DEA, Model Risk, RAGAS all first-time; allow buffer
Pioneer 2 — Customer Verification Agent	WP-007	3	7	H1	Medium — builds on Pioneer 1 learnings; pattern established
Pioneer 3 — Advisory Rebalancing Agent [Programme Risk Gate]	WP-008	2	6	H2	Medium — MiFID II HITL adds complexity; suitability rules in KB-003
Pioneer 4 — IT Incident Response Agent	WP-009	2	5	H2	Low — operational domain; fewer regulatory constraints
Pioneer 5 — Model Monitoring & Compliance Agent	WP-010	3	6	H2	Medium — DORA reporting integration; cross-domain data access
Self-service portal, agent marketplace, agent health dashboard	WP-011	3	8	H2	Medium — developer experience is subjective; UX validation needed
EU AI Act compliance pack + DEA template library + DORA automation	WP-012	2	5	H2	High — regulatory interpretation risk; regulatory counsel required

Solution Building Blocks — SBB-APEX-001

DOC REF: SBB-APEX-001 | APPROVED

SBB	Decision	Cloud-Native Solution	AI-DLC Role in Delivery
Agent Orchestration Engine	EVALUATE OSS preferred	LangGraph / AutoGen / CrewAI — evaluated in WP-002 spike; abstraction layer prevents lock-in	AI-DLC Construction generates agent YAML, tool definitions, and action group config
Foundation Models	BUY via provider-agnostic adapter	Provider evaluated per task; no single-vendor lock-in; LiteLLM-style adapter library in all agent code	AI-DLC Construction: AI selects model per task type from pre-approved catalogue updated quarterly
Vector Store	OSS preferred	Weaviate / Qdrant / Milvus — Embedding Compatibility Contract applies	AI-DLC Construction generates index schema, query patterns, and RAGAS test suite
Agent Control Plane	BUILD	Python FastAPI; Kubernetes; 6 services (ALM, Budget, Onboarding, Chargeback, Notification, Compliance)	AI-DLC builds each service in 1 bolt; L1–L5 verification on all IaC and code

SBB	Decision	Cloud-Native Solution	AI-DLC Role in Delivery
DEA Generation Service	BUILD	Python event-driven service; Kubernetes; immutable store	AI-DLC builds DEA assembler in 1 bolt; Compliance Lead reviews output against EU AI Act Art.13
Agent Gateway	BUILD on OSS	Envoy proxy + custom plugin OR APISIX + custom plugin	AI-DLC generates gateway config from agent registry schema; property tests validate enforcement rules
Developer Portal	BUILD	React SPA + API Gateway; Amplify or equivalent CDN	AI-DLC scaffolds UI components in Mob Construction in 1 bolt; UX validated with domain dev teams
Human Review Portal	BUILD on open framework	React + Temporal / Camunda workflow engine	AI-DLC generates workflow JSON definitions; custom UI mob-built in 1 bolt
LGTM Observability	OSS	OpenTelemetry + Prometheus + Grafana + Loki + Tempo (full LGTM stack)	AI-DLC generates dashboards-as-code (Grafana JSON) from agent metric schema definition
AI Safety Layer	OSS	NeMo Guardrails or Guardrails AI; configured per agent EU AI Act risk tier	AI-DLC generates guardrail rule sets; red-team adversarial tests AI-generated quarterly
IaC Platform	OSS (REUSE)	Terraform / OpenTofu + Argo CD (GitOps); Terratest for property tests	AI-DLC extends existing pipeline; Terratest property tests are mandatory (L3 verification)
Compliance Reporter	BUILD on OSS	Python pipeline + Apache Superset or Metabase	AI-DLC generates EU AI Act and DORA report templates; DORA change record auto-writer component added

5-Tier Governance Model — CAF-APEX-001

DOC REF: CAF-APEX-001 | APPROVED

Tier	Scope	Method	SLA	Bypassable?
T1 — Auto	IaC L1 scan; RAGAS score; DEA completeness; tag compliance; budget alert	CI pipeline + Kubernetes admission controllers + scheduled validation jobs	Synchronous	NO — pipeline physically blocks; no human override path
T2 — Architect	Pattern conformance; new data flow; data classification audit	Automated flag → architect on-call (Slack/Teams) within 4-hour SLA	4 business hours	NO — deployment blocked until architect approves
T3 — ARB	New SBB; significant deviation; agent topology change; debt threshold breach	Weekly ARB session; async for urgent items between sessions	Weekly	Exception ONLY: Group CTO written approval + compliance register entry; silent bypass is not possible
T4 — Reg-Gate	High-Risk AI (EU AI Act Art.6); DORA-reportable change; any C4-data flow change	Compliance Lead; no delegation; Regulatory Affairs Lead engaged	48 hours	NO — hardcoded in CI pipeline; no exception pathway exists
T5 — Exec	Budget above approved threshold; strategic architecture change; programme risk escalation	Monthly Steering Committee; emergency session within 48 hrs for DORA incidents	Monthly	NO — requires Steering Committee quorum

Change Intelligence Process — ACHG-APEX-001

DOC REF: ACHG-APEX-001 | APPROVED – DORA Change Framing Applied to All AI Evolution

Change Type	AI Monitoring	Human Decision Authority	DORA Framing	Illustrative Example
T1 — Technology-driven	Model release channels monitored; OSS project feeds; benchmark auto-run vs. GlobalCorp eval suite	Enterprise Architect reviews impact; ARB approves; T4 Reg-Gate if High-Risk agent affected	Formal change event regardless of model release frequency	Open-weight model outperforms current by 18%; ARB approves upgrade in 3 days with RAGAS regression evidence

Change Type	AI Monitoring	Human Decision Authority	DORA Framing	Illustrative Example
T2 — Regulatory-driven	Regulatory feed parsed daily by compliance-monitoring agent; impact classified against APEX control inventory	Compliance Lead decides; Regulatory Affairs Lead engaged for ambiguous interpretation	Regulatory change triggers controlled response plan; not reactive patching	EU AI Act Art.6 enforcement date reached; APEX intake auto-classification activated; DORA evidence pack updated same day
T3 — Business-driven	Onboarding request submitted via self-service portal; intake workflow triggered automatically	ARB intake decision within 5 business days; architecture review within 10 business days	Standard change; ARB record created; DORA log entry if affects Tier-1 service	New Procurement Division requests APEX onboarding for contract analysis use case
T4 — Simplification	Usage analytics flags agents below utilisation threshold for 3 consecutive months; auto-review prompt generated	Domain owner decides: decommission or document justification for retention	Decommission = formal change event; IaC teardown + DEA archive + compliance record	Pioneer 1 v1.2 superseded by v2.0; v1.2 parallel-run for 4 weeks then decommissioned with full evidence trail

REQ MGMT

Requirements Management — Continuous

Living ARS — 162 total requirements across all phases

Requirements Management is the continuous central process of the TOGAF ADM. AI-DLC's Semantic Context Building and persistent context storage transform this from a periodic document into a living, AI-maintained traceability system with a human review gate before every deployment.

DOC REF: ARS-APEX-001 | APPROVED | 162 total requirements

Req ID	Category	Statement	Priorit y	Source	Status
BR-001	Business	Platform supports at minimum 120 concurrent agent interactions per region under normal load	Must	Capacity model	In Design
BR-003	Business	All agent decisions logged with full context and queryable for a minimum of 7 years	Must	DORA Art.11	In Design
BR-006	Business	EU AI Act risk tier automatically classified for every agent at the point of intake	Must	EU AI Act Art.6	In Design
BR-007	Business	Productivity gains tracked quarterly against actuals; reported to Steering Committee	Must	BP-01	In Design
DR-001	Data	Personal data never persisted in conversation logs, session memory, or observability traces	Must	GDPR Art.5	In Design
DR-002	Data	Embedding model ID and version stored in data catalog for every KB; RAGAS gate on any upgrade	Must	DP-05	In Design
DR-003	Data	Previous embedding index retained for 90 days post-upgrade to support rollback	Must	DP-05	In Design
AR-001	Application	Agent Control Plane API achieves p99 response time below 500ms under 500 RPS	Must	SLA Policy	In Design
AR-002	Application	DEA generation service produces complete DEA within 5 seconds; stored immutably for 7 years	Must	TP-06	In Design
AR-003	Application	Agent Gateway enforces registry lookup; unregistered agents receive 403 at gateway	Must	BP-03	In Design
TR-001	Technology	All AI model invocations travel exclusively over private network paths; no public internet egress from agent pods	Must	TP-01 / TP-02	In Design
TR-002	Technology	All AI-generated IaC passes L1–L5 layered verification; L2 explanation artefact retained 12 months	Must	TP-05	In Design
SR-001	Security	Customer-managed encryption keys rotated on a maximum 365-day cycle; rotation enforced by automated check	Must	GDPR Art.32	In Design
SR-002	Security	Red-team adversarial testing of every agent completed and documented before production deployment approval	Must	AI Risk Policy / OWASP LLM	In Review
CR-001	Compliance	EU AI Act risk classification documented and stored in agent registry for every agent in all environments	Must	EU AI Act Art.9	In Design
CR-002	Compliance	Model explainability method (SHAP, LIME, or equivalent) documented; evidence artefact generated before production	Must	EU AI Act Art.13	In Design
CR-003	Compliance	DORA 72-hour ICT incident notification workflow automated and tested quarterly under simulated conditions	Must	DORA Art.17	In Review
CR-004	Compliance	Every High-Risk AI agent produces a DEA for every regulated decision; DEA queryable by compliance team	Must	EU AI Act Art.13 / DEA	In Design

Req ID	Category	Statement	Priorit y	Source	Status
CR-005	Compliance	Every AI model update, KB refresh, and agent topology change recorded as a formal DORA change event with ARB reference	Must	DORA Art.11	In Design

APPENDIX A

Regulatory Cross-Reference Matrix

APEX compliance posture across all applicable regulatory frameworks

Regulation	Key Obligations	APEX Architecture Control	Evidence Artefact
EU AI Act 2024/1689	Art.6 (High-Risk classification), Art.9 (Risk Management System), Art.13 (Transparency — addressed by DEA pattern), Art.17 (Quality Management)	EU AI Act risk classifier at intake (WP-012); DEA pattern for all High-Risk agents; RAGAS gate; Model Risk opinion; DEA export API; explainability artefact in CI/CD	DEA per regulated decision; EU AI Act compliance pack (WP-012); Model Risk Opinions; CI/CD explainability reports
DORA (EU) 2025	Art.9 (ICT security policy), Art.11 (ICT risk management — all AI changes = controlled change events), Art.17 (72-hr incident notification), Art.26 (TLPT)	Zero-trust security architecture; DORA change framing protocol (every AI change = formal change event with ARB record); Incident Agent (Pioneer 4); automated 72-hr notification workflow; quarterly red-team testing	DORA Evidence Pack (quarterly); DORA change records per AI update; incident runbook test results; red-team findings
GDPR / UK GDPR	Art.5 (data minimisation), Art.25 (privacy by design — Phase Boundary Receipts enforce this), Art.32 (appropriate security), Art.46 (cross-border transfer controls)	PII ANONYMIZE in guardrail layer; data residency enforced via NetworkPolicy; AES-256 CMK encryption; agent memory TTL; Phase Boundary Receipt requires DPO sign-off before C4 data flows go to production	Article 30 records of processing; Data Flow Maps per Pioneer agent; Guardrail configuration evidence; DPO sign-off artefacts
Internal Model Risk (aligned SR 11-7)	Model validation before production; documentation of assumptions; ongoing performance monitoring; independent review	Model Risk opinion gate (T4 Reg-Gate) in CI/CD; explainability artefact (SHAP/LIME); RAG retrieval explanation in DEA; Model Monitoring Agent (Pioneer 5) for drift detection	Model Risk Opinions per Pioneer agent; Explainability reports (SHAP/LIME per model); Drift monitoring dashboards (Pioneer 5)
BCBS 239	Data lineage for risk aggregation; accuracy, completeness, and timeliness of risk data reporting	Data catalog lineage integration; RAG retrieval log traces every agent decision to source data; DEA RAG section provides full lineage chain to regulators and auditors	Data catalog lineage reports; RAG retrieval log exports (7-year retention); DEA RAG retrieval section per regulated decision
MiFID II Art.27	Best execution obligations; suitability assessment for investment advice; record-keeping for advisory decisions	Human-in-the-Loop mandatory for all investment advisory outputs; HITL outcome written to DEA; KB-003 contains suitability rules; full session and decision logging	DEA with HITL section; Suitability assessment records; Session logs per advisory interaction; Agent audit trail

APPENDIX B

Cloud-Native Service Mapping

OSS reference implementations and cloud-provider equivalents

All APEX capabilities are specified using open-source and open-standard references. Cloud-provider managed equivalents are listed for reference. Provider selection occurs in Phase F based on commercial evaluation, existing contracts, and regional service availability.

APEX Capability	OSS / Open-Standard	Provider A (AWS)	Provider B (GCP)	Provider C (Azure)
Container Orchestration	Kubernetes (CNCF)	EKS	GKE	AKS
Serverless Functions	Knative / OpenFaaS	AWS Lambda	Cloud Functions	Azure Functions
Agent Orchestration	LangGraph / AutoGen / CrewAI	Bedrock Agents	Vertex AI Agents	Azure AI Foundry Agents
Foundation Model API	Ollama (local) / LiteLLM adapter	Bedrock (Claude, Titan)	Vertex AI (Gemini, Claude)	Azure OpenAI (GPT-4o, o1)
Vector Store	Weaviate / Qdrant / Milvus	OpenSearch Serverless	AlloyDB + pgvector	Azure AI Search
Object Storage	MinIO (S3-compatible API)	S3	Cloud Storage	Azure Blob Storage
Managed Relational DB	PostgreSQL (self-managed on K8s)	Aurora PostgreSQL	Cloud SQL PostgreSQL	Azure DB for PostgreSQL
Event Streaming	Apache Kafka / Redpanda	Kinesis / MSK	Pub/Sub / Dataflow	Azure Event Hubs / Service Bus
Workflow Engine	Temporal / Argo Workflows	Step Functions	Cloud Workflows	Azure Durable Functions
Secrets Management	HashiCorp Vault	AWS Secrets Manager + KMS	Secret Manager + Cloud KMS	Azure Key Vault
Metrics	Prometheus (CNCF)	CloudWatch Metrics	Cloud Monitoring	Azure Monitor
Tracing	Jaeger / Grafana Tempo (OTEL)	AWS X-Ray	Cloud Trace	Azure Application Insights
Logging	Loki / OpenSearch	CloudWatch Logs	Cloud Logging	Azure Monitor Logs
Dashboards	Grafana (CNCF)	CloudWatch Dashboards	Cloud Monitoring Dashboards	Azure Dashboards
GitOps / CD	Argo CD / Flux (CNCF)	CodePipeline	Cloud Deploy	Azure DevOps Pipelines
Service Mesh	Istio / Linkerd (CNCF)	AWS App Mesh	Traffic Director	Azure Istio add-on
ML Experiment Tracking	MLflow	SageMaker Experiments	Vertex AI Experiments	Azure ML Experiments
AI Safety / Guardrails	NeMo Guardrails / Guardrails AI	Bedrock Guardrails	Vertex AI Safety Filters	Azure AI Content Safety
IaC	Terraform / OpenTofu + Terratest	AWS CDK / CloudFormation	Deployment Manager	Bicep / ARM Templates
Progressive Delivery	Argo Rollouts / Flagger	CodeDeploy	Cloud Deploy canary	Azure Deployment Strategies
Agent Gateway	Envoy + plugin / APISIX + plugin	API Gateway + Lambda authoriser	Apigee + custom policy	API Management + policy
IaC Security (L1)	Checkov + Semgrep + Trivy	CodeGuru + Inspector	Artifact Registry scanning	Defender for DevOps
Runtime Security	Falco (CNCF)	GuardDuty	Security Command Center	Defender for Containers

APEX Capability	OSS / Open-Standard	Provider A (AWS)	Provider B (GCP)	Provider C (Azure)
DEA Store	PostgreSQL + object storage (WORM) + Athena	S3 Object Lock + Athena	Cloud Storage (locked) + BigQuery	Azure Immutable Blob + Synapse

APPENDIX C

Glossary

Complete reference — TOGAF 10, AI-DLC, cloud-native, team, and regulatory

ADM	Architecture Development Method — the iterative process at the core of TOGAF 10; all phases can run concurrently or be scoped independently based on programme needs.
Adaptive Capability	A business capability whose behaviour changes continuously through AI learning; requires dynamic governance distinct from static capabilities, including DORA change framing for every evolution.
Agent Gateway	Platform component that enforces agent registry lookup, budget hard-throttle at 110%, dependency-map logging, and blocks unregistered agents (403 response) before any model invocation.
Agent Health Score	Monthly composite score per agent: usage volume, error rate, cost-per-decision, human escalation rate; agents below threshold flagged for decommission review.
AI-DLC	AI-Driven Development Lifecycle — AWS methodology (July 2025) positioning AI as central collaborator; three phases: Inception (Mob Elaboration), Construction (Mob Construction), Operations.
APEX	AI Platform of Platforms — GlobalCorp's programme to build a shared, governed, cloud-native AI agent platform eliminating siloed AI tool sprawl.
ARB	Architecture Review Board — governance body that approves architecture decisions, manages conformance, and maintains architecture principles; meets weekly under AI-DLC cadence.
Bolt	AI-DLC development cycle measured in hours to days; replaces the Agile sprint. APEX planning baseline: 1 bolt ≈ 1–2 weeks for Pioneer Domain agents including governance gates.
CNCF	Cloud Native Computing Foundation — open-source foundation hosting Kubernetes, Prometheus, OpenTelemetry, Argo CD, Falco, Istio, and other APEX-standard technologies.
Data Maturity Gate	Mandatory pre-Phase B assessment: 5 checks covering documentation currency, process coverage, data lineage completeness, terminology consistency, and architecture debt baseline.
DEA	Decision Explanation Artefact — structured record of every component of a regulated AI decision: routing, RAG retrieval, model invocation, guardrail evaluation, output, and HITL outcome (if applicable). 7-year retention. Addresses EU AI Act Art.13.
DORA	EU Digital Operational Resilience Act — requires ICT risk management, 72-hour major incident notification, and TLPT. APEX treats all AI evolution as formal DORA change events.
Embedding Compatibility Contract	Governance rule requiring that any embedding model upgrade triggers full re-indexing of all dependent KBs and must pass a RAGAS regression gate before production activation.
EU AI Act	EU Regulation 2024/1689 — classifies AI systems by risk; mandates transparency (Art.13, addressed by DEA), human oversight, explainability, and quality management for High-Risk systems.
GitOps	Operational model where infrastructure and application desired state is declared in Git and continuously reconciled by an automated operator (Argo CD or Flux).
Guardrails	Content safety layer intercepting agent inputs/outputs; enforces PII anonymisation, topic filtering, and grounding checks; grounding threshold 0.75 non-negotiable.
HITL	Human-in-the-Loop — architecture pattern routing agent outputs above a risk threshold to a human reviewer; HITL outcome is written into the DEA.
Layered Verification Model	Five-layer IaC and code verification: L1 static analysis, L2 AI explanation artefact, L3 property tests, L4 canary deploy, L5 drift detection; replaces single-person hero review.
LGTM Stack	Loki + Grafana + Tempo + Mimir (or Prometheus) — the open-source observability stack standard for all APEX agents.
Mob Construction	AI-DLC ritual where the full cross-functional team builds solution components — architecture, IaC, code, tests — in a shared real-time session with AI as collaborator.
Mob Elaboration	AI-DLC ritual transforming business intent into requirements, capability maps, and architecture units; must be preceded by a political context capture session to capture what AI cannot.
OpenTofu	Open-source fork of Terraform under Linux Foundation / CNCF; OSS alternative without BSL licence constraints; APEX preferred IaC standard.
Phase Boundary Receipt	Formal evidence artefact confirming minimum sign-off criteria for a TOGAF phase boundary are met; DPO sign-off required before C4 data flows transition from Phase C to Phase D production.
RAGAS	RAG Assessment framework — measures faithfulness, answer relevance, context recall, and context precision; used as the regression gate for all KB and embedding model changes.
SAW	Statement of Architecture Work — TOGAF artefact formally initiating an architecture engagement; defines scope, constraints, resources, milestones, and is signed by executive sponsor.

SBB	Solution Building Block — specific implementation of an Architecture Building Block; defines what is built, bought, or reused for a given platform capability.
Semantic Context	AI-DLC concept: rich, structured context provided to AI coding agents through steering files so they apply GlobalCorp-specific standards automatically and consistently.
TLPT	Threat-Led Penetration Testing — DORA Art.26 obligation for major entities; GlobalCorp APEX conducts TLPT quarterly for the agent gateway and core platform components.
Unit of Work	AI-DLC equivalent of an Agile Epic; decomposed into bolts. APEX planning baseline: 1 Unit of Work for a Pioneer Domain agent = 3 bolts ≈ 6–8 calendar weeks including governance gates.
Zero-Trust	Security model where no identity or network location is inherently trusted; every request is authenticated, authorised, and logged regardless of origin; enforced by service mesh mTLS.

Document Control

Version	Issue Date	Authors	Description	Approved By
1.0	January 2025	Enterprise Architecture Team	Initial release — TOGAF 10 ADM phases A–D	Architecture Review Board
2.0	March 2025	Enterprise Architecture Team	Full ADM coverage; Architecture Contracts; Governance model added	Group CTO
3.0	April 2026	Enterprise Architecture Team	Cloud-native edition (vendor-agnostic); AI-DLC integrated; table formatting improved	Group CTO
4.0 — FINAL	April 2026	Enterprise Architecture Team	Final edition: team structure added (org model, core team register, RACI: Agent Lifecycle, RACI: Platform Governance, RACI: Regulatory, Responsibility Assignment Matrix, Stakeholder Map, Team Interaction Map); DEA service; Agent Gateway; Embedding Compatibility Contract; Data Maturity Gate; Phase Boundary Receipts; DORA change framing; Layered Verification Model; Agent Sprawl governance; 19 requirements; 24-entry Cloud-Native Service Mapping; full glossary	Group CTO

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